

## WAVE HEIGHT

The further a wave travels, the bigger it can get. A gale (strong wind) blowing over a small lake will create only small waves, but a wind of the same strength blowing over an ocean can create waves that are over 33 ft (10 m) high. The biggest waves build up in the Southern Ocean, where there is no land to stop them sweeping around Antarctica, driven by the westerly wind.

### ▼ ATLANTIC STORM

*Howling winds blow spray off the crests of the giant waves threatening this fishing boat.*

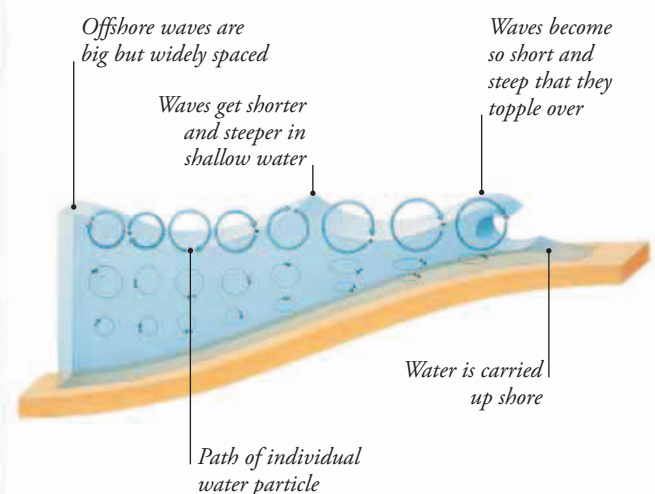


## ROGUE WAVES

Out at sea, regular swells can be very high without being particularly dangerous. But if two swells come together, they can clash to form colossal rogue waves more than 66 ft (20 m) high. These also form where a series of storm waves meet a strong opposing current. Such waves can wash right over big ships, and may even sink them.

## BREAKERS

As waves approach the shore and move into shallower water, they become shorter and steeper. This makes each wave more top-heavy, until finally its crest topples forward in a foaming mass of water called a breaker. The steeper the shore slopes up from deep water, the more dramatically the waves break, hurling water up the beach.



**WOW!**

In 1995, the ocean liner *Queen Elizabeth 2* was hit by a rogue wave during a hurricane in the Atlantic. The wave was about 95 ft (29 m) high.

